

ACCTG 211H — Mechanics Drill Set

Week 11 — Cost Behavior and Cost-Volume-Profit Analysis

i Graded for **completion**. Work each drill by hand, then check the answer key (posted separately) and re-do any you missed before the quiz.

Drill Set

Quick, single-skill reps to build mechanical fluency before the week's first class. Graded for **completion** — work each one by hand, then check it against the key that follows and re-do any you missed.

1. (*High-low method.*) At 10,000 units of activity total cost is \$80,000; at 6,000 units total cost is \$56,000. Compute (a) the variable cost per unit and (b) total fixed cost.
2. (*Contribution margin.*) A product sells for \$40 and has a variable cost of \$25 per unit. Compute (a) the unit contribution margin and (b) the contribution-margin ratio.
3. (*Break-even in units.*) Fixed costs are \$120,000 and the unit contribution margin is \$15. Compute the break-even point in units.
4. (*Break-even in dollars.*) Fixed costs are \$120,000 and the contribution-margin ratio is 37.5%. Compute break-even sales dollars.
5. (*Target profit.*) Using a unit contribution margin of \$15 and fixed costs of \$120,000, compute the number of units required to earn a target operating profit of \$45,000.
6. (*Margin of safety.*) Budgeted sales are \$500,000 and break-even sales are \$320,000. Compute the margin of safety in dollars and as a percentage of sales.
7. (*Contribution-margin income statement.*) A firm sells 11,000 units at \$40 each; variable cost is \$25 per unit; fixed costs are \$120,000. Prepare a contribution-margin income statement (sales, variable costs, contribution margin, fixed costs, operating income).
8. (*Operating leverage.*) Using the contribution margin (\$165,000) and operating income (\$45,000) from Drill 7, compute the degree of operating leverage, and predict the percentage change in operating income if sales rise 10%.